



Measurements of electroweak $W^{\pm}Z$ boson pair production in association with two jets in pp collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector

Phenomenology of the Standard Model of Elementary Particles

Master's Degree in Physics - Elementary Particles and

Astroparticles

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Probing the EW Sector and BSM Physics with $W^\pm Zjj$

1 Introduction

- **Dataset:** 140 fb^{-1} of pp collisions at $\sqrt{s} = 13 \text{ TeV}$ (2015–2018).
- **Target:** Fully leptonic decays of $W^\pm Zjj$ production.
- **Measurements:** Integrated & differential fiducial cross-sections (σ_{EW} and σ_{strong}).
- **Goal:** Isolate Vector Boson Scattering (VBS).
- **New Physics:** EFT limits on Anomalous Quartic Gauge Couplings (aQGC).



The ATLAS Detector

1 Introduction

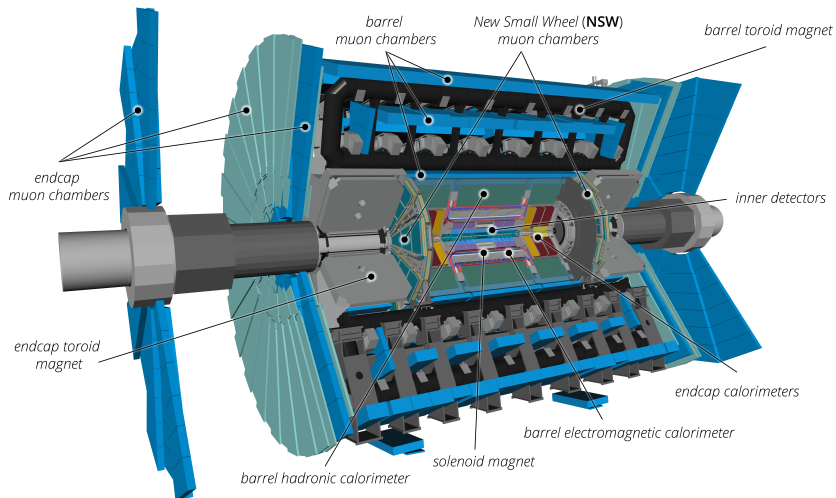




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Gauge Boson Self-Interactions

2 Theoretical Framework

- The Electroweak (EW) sector of the Standard Model is based on the **non-abelian** gauge symmetry $SU(2)_L \times U(1)_Y$.
- Unlike QED, the generators of $SU(2)_L$ do not commute. As a consequence, the physical vector bosons ($W^\pm, Z, \gamma$) self-interact.
- This structure directly dictates the existence of **Triple** (TGC) and **Quartic** (QGC) Gauge Couplings:

$$\mathcal{L}_{\text{TGC}} = \frac{1}{2} g \epsilon_{abc} W_b^\mu W_c^\nu (\partial_\mu W_{a\nu} - \partial_\nu W_{a\mu}) \quad \mathcal{L}_{\text{QGC}} = \frac{1}{4} g^2 \epsilon_{abc} \epsilon_{ars} W_b^\mu W_c^\nu W_{r\mu} W_{s\nu}$$

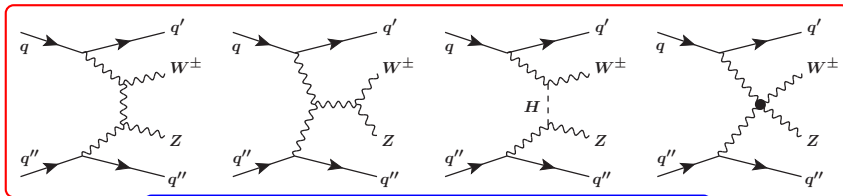
Probe these vertices to test the SM rigidity and search for anomalous couplings (aQGC) arising from New Physics.



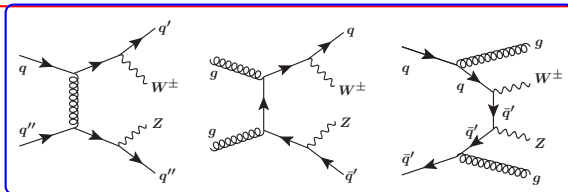
Experimental Signature: Vector Boson Scattering

2 Theoretical Framework

Vector Boson Scattering (VBS) $VV \rightarrow VV$ with $V = W/Z/\gamma$, is a key process with which to probe the $SU(2)_L \times U(1)_Y$ gauge symmetry of the **electroweak** (EW) theory that determines the **self-couplings** of vector bosons. To study these couplings, we exploit the purely electroweak production of $W^\pm Zjj$.



$W^\pm Zjj$ -EW



$W^\pm Zjj$ -QCD



A little look at the Effective Field Theory (EFT)

2 Theoretical Framework

The $W^\pm Zjj$ -EW production can be sensitive to effects beyond the SM affecting the quartic interactions of weak bosons. The measured $W^\pm Zjj$ events are therefore used to search for anomalous quartic gauge couplings (aQGC) using an EFT framework. In this model, the Lagrangian of the standard model is extended into an efficient Lagrangian by adding higher-order operators and their respective Wilson coefficients as follows:

$$\mathcal{L}_{\text{eff}} = \mathcal{L}_{\text{SM}} + \sum_i \frac{f_i^{(6)}}{\Lambda^2} O_i^{(6)} + \sum_j \frac{f_j^{(8)}}{\Lambda^4} O_j^{(8)} + \dots$$

Where:

- $O_{i,j}^{(6),(8)}$: are dimension-6 and dimension-8 operators;
- $f_i^{(6)}$ e $f_j^{(8)}$: dimensionless couplings
- Λ : is the energy scale of the new processes



A little look at the Effective Field Theory (EFT)

2 Theoretical Framework

The squared scattering amplitude of the effective field theory prediction for $W^\pm Z jj$ production can be written as:

$$\left| A_{\text{SM}} + \sum_i c_i A_i \right|^2 = |A_{\text{SM}}|^2 + \sum_i c_i 2 \text{Re}(A_{\text{SM}}^* A_i) + \sum_i c_i^2 |A_i|^2 + \sum_{ij, i \neq j} c_i c_j 2 \text{Re}(A_i A_j^*)$$

- $c_i = f_j^{(8)} / \Lambda^4$, A_{SM} is the SM scattering amplitude;
- $2 \text{Re}(A_{\text{SM}}^* A_i)$ is the amplitude of the interference term between the SM and the dimension-8 operators;
- $\sum_i c_i^2 |A_i|^2$ is the pure dimension-8 contribution, and $\sum_{ij, i \neq j} c_i c_j 2 \text{Re}(A_i A_j^*)$ is the amplitude of interferences between two dimension-8 operators, called cross terms.



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Fiducial Phase Space for Leptons

3 Fiducial Phase Space and MC Simulation

The $W^\pm Z jj$ cross-sections are measured in a fiducial phase space. Leptons produced in the decay of a hadron, a τ -lepton, or their descendants are not considered in the definition of the fiducial phase space and are treated as background.

Category	Kinematic Variable	Requirement
Z leptons	Transverse momentum ($p_T^{\ell_Z}$)	$> 15 \text{ GeV}$
W lepton	Transverse momentum ($p_T^{\ell_W}$)	$> 20 \text{ GeV}$
All leptons	Pseudorapidity ($ \eta_\ell $)	< 2.5
Z boson	Invariant mass window ($ m_{\ell^+\ell^-} - m_Z^{\text{PDG}} $)	$< 10 \text{ GeV}$
W boson	Transverse mass (m_T^W)	$> 30 \text{ GeV}$
Angular isolation	Distance between Z leptons ($\Delta R(\ell_{Z1}, \ell_{Z2})$)	> 0.2
Angular isolation	Distance between W and Z leptons ($\Delta R(\ell_W, \ell_Z)$)	> 0.3



Fiducial Phase Space for Jets

3 Fiducial Phase Space and MC Simulation

In addition, at least two jets are required, reconstructed using the anti- k_t algorithm with radius parameter $R = 0.4$. Hard processes with a b -quark in the initial or final states, such as tZj production, are not considered as part of the $W^\pm Zjj$ -EW events.

Category	Kinematic Variable	Requirement
All jets	Transverse momentum (p_T^j)	$> 40 \text{ GeV}$
All jets	Pseudorapidity ($ \eta_j $)	< 4.5
Angular isolation	Distance between jets and leptons ($\Delta R(j, \ell)$)	> 0.3
Tagging jets	Opposite hemispheres ($\eta_{j1} \cdot \eta_{j2}$)	< 0
Tagging jets	Invariant mass (m_{jj})	$> 500 \text{ GeV}$
Jet Veto		
b -quark jets	Absence of b -jets in initial/final state	$N_{b\text{-jet}} = 0$



How to eliminate leptons ambiguity?

3 Fiducial Phase Space and MC Simulation

- Leptons and final-state neutrons that do not originate from hadron or τ -lepton decays, are matched to the $W^\pm$ and Z boson decay products using an algorithmic approach, called the **resonant shape algorithm**.
- This algorithm is based on the value of an estimator expressing the product of the nominal line-shapes of the W and Z resonances:

$$P = \left| \frac{1}{m_{(\ell^+, \ell^-)}^2 - (m_Z^{\text{PDG}})^2 + i \Gamma_Z^{\text{PDG}} m_Z^{\text{PDG}}} \right|^2 \times \left| \frac{1}{m_{(\ell', \nu_{\ell'})}^2 - (m_W^{\text{PDG}})^2 + i \Gamma_W^{\text{PDG}} m_W^{\text{PDG}}} \right|^2$$



Monte Carlo Simulation of the Signal

3 Fiducial Phase Space and MC Simulation

The simulation of $W^\pm Zjj$ events was primarily generated using the MadGraph5_aMC@NL02.6.5 MC event generator, interfaced with Pythia8.240. The simulation was divided into distinct components:

- **Electroweak production $W^\pm Zjj$ – EW:** The actual signal. It is calculated at leading order (LO), considering purely electroweak processes (order α_{EW}^6).
- **QCD production $W^\pm Zjj$ – QCD:** Considered a background for the VBS analysis. This component is calculated at next-to-leading order (NLO) in QCD.
- **Interference $W^\pm Zjj$ – INT:** Quantifies the quantum interference between the EW and QCD production. It is calculated at LO and yields a positive contribution of 6% in the fiducial region, reaching up to 12% for events with three or more jets.



Monte Carlo Simulation of the Background

3 Fiducial Phase Space and MC Simulation

In addition to the $W^{\pm}Zjj$ – QCD production, there are numerous other Standard Model processes that can mimic the signal in the detector. These backgrounds fall into two categories based on how they are estimated:

- **Backgrounds with *prompt (real)* leptons:** Processes such as the production of ZZ pairs, WW , tribosons (VVV), or top quarks associated with vector bosons ($t\bar{t}V$, tZj). Their contribution is evaluated entirely using Monte Carlo simulations.
- **Backgrounds with *fake* leptons:** Events where a hadronic jet is misidentified as a lepton by the detector. Being an instrumental artifact that is complex to simulate accurately, this background is estimated directly using data-driven techniques.



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Thank you for listening!

Any questions?